

Unit

2

Lesson

3



Innovators in Action.

Code, Create, Conquer!

Time Frame: [90 min]

Ready, Set, Learn!

You Need to Know

- Why separating wet and dry waste is important for recycling and waste management
- How a moisture sensor works to detect wet and dry waste.
- Differentiates between Analog and Digital components.
- The importance of planning, measuring, and safety when building a project.

Challenge questions of the lesson.

1. Why is it important to separate wet and dry waste before throwing it into a smart bin?
2. How does a moisture sensor “know” if something is wet or dry, and what kind of data does it send to the Arduino?
3. Imagine you're building a smart bin with a sensor and motor inside. What are three things you must plan?

SEL Relation

- **Community Awareness & Problem Solving (Social Awareness & Responsible Decision-Making):** Students will consider real-world problems related to waste management, hygiene, and environmental responsibility within their community. They will use their understanding of these issues to guide their design choices for a more efficient and user-friendly smart bin, fostering a sense of civic responsibility.
- **Collaboration and Communication (Relationship Skills):** Students will work effectively together (if in groups) to brainstorm design ideas for the bin's structure and electronic integration, share circuit concepts, troubleshoot technical challenges, and provide constructive feedback, developing strong communication and teamwork skills throughout the building process.
- **Innovation and Persistence (Self-Awareness & Self-Management):** Students will face the complex challenge of designing and constructing a functional automated bin, requiring them to think creatively to integrate components, analyze constraints, and persist through troubleshooting. This process builds self-awareness of their problem-solving abilities and enhances their self-management when encountering difficulties.
- **Applied Responsibility & Safety (Responsible Decision-Making):** The lesson inherently encourages students to make responsible choices regarding material selection, accurate measurement, and safe tool handling during construction. This fosters a sense of responsibility for their work and the safety of themselves and their peers within the classroom environment.

Engage



Teacher Role:

- **Initiator & Facilitator:** Begin by posing the challenge of sorting wet and dry waste, mirroring Mrs. Sara's dialogue: "Do you remember when we talked about the difference between types of waste? A banana peel is wet, while a piece of paper is dry. Our smart bin has to figure that out on its own to sort things correctly. How do you think it can do that?". Guide the discussion, ensuring all students understand the problem and the initial ideas for how the bin could distinguish and sort.
- **Questioner & Clarifier:** Pose open-ended questions to encourage students to think about how a machine can "see" the difference, leading them to identify the appropriate sensor as in the dialogue ("Maybe with a sensor?", "Does anyone know the name of the sensor that will act like the bin's 'eyes' to tell if something is wet or dry?", "Is it the moisture sensor?", "Yes, well done!"). Address any vocabulary or process-related questions that come up during the discussion, especially terms like "moisture sensor," "signal," "Arduino," and "command."
- **Task Setter:** Introduce the specific problem-listing task: "Now, Let's learn how the sensor works. List out at least three problems that could happen if food scraps and paper or plastic are all thrown into the same bin".
- **Connector:** Help students connect the idea of a sensor (like the moisture sensor) getting information, that information going to the "brain" (Arduino), and the "brain" then telling a "muscle" (motor) what to do to sort the waste.
- **Motivator:** Build enthusiasm for the upcoming hands-on activities focused on assembling and testing the smart bin's actual sorting mechanism, framing it as a solution to the problems they will identify.

Expected From Students:

- **Active Participation in Dialogue:** Students will listen attentively and actively engage in the discussion about distinguishing wet/dry waste, contributing ideas and identifying the moisture sensor when prompted.
- **Problem Identification:** Students will list out at least three problems that could happen if food scraps and paper or plastic are all thrown into the same bin.
- **Conceptual Understanding:** Students will grasp the idea that a sensor can detect properties (like moisture), send a signal, and that an "Arduino" can interpret this signal to control a "motor" for sorting.
- **Active Listening:** Students will listen attentively to the discussion, identifying the core problem of distinguishing wet/dry waste and following the proposed solution steps.

- **Initial Problem-Solving:** Students will begin to think about the logical sequence required for automated waste sorting: sensing → processing → acting, and the negative consequences of unsorted waste.

Parts of the books included.



Story



- **Hook with a Real-World Problem:** Begin by saying: "Have you ever seen someone try to put a banana peel and a piece of paper into the same recycling bin? What happens if they get all mixed up? Why is it important to keep wet waste and dry waste separate?" Let students respond, guiding them towards environmental and practical reasons (e.g., wet waste contaminates recyclables, smelly, attracts pests).
- **Engage with the Challenge Dialogue:** Say: "Today, we're going to step into a classroom conversation just like ours, where Mrs. Sara, Adam, and Laila face this very challenge for our smart bin. Let's find out how they start solving it!" Read the provided dialogue aloud using expressive voices for each character. Emphasize Mrs. Sara's guiding questions and Laila's excited "Aha!" moment.
- **Dialogue:** Mrs. Sara: Do you remember when we talked about the difference between types of waste? A banana peel is wet, while a piece of paper is dry. Our smart bin has to figure that out on its own to sort things correctly. How do you think it can do that? Adam: Maybe with a sensor? Mrs. Sara: Great thinking, Adam! Does anyone know the name of the sensor that will act like the bin's "eyes" to tell if something is wet or dry? Laila: Is it the moisture sensor? Mrs. Sara: Yes, well done! Now, Let's learn how the sensor works.
- **Make the Problem Personal:** After reading the dialogue, ask: "Based on what Mrs. Sara and the students discussed, what's the big problem our smart bin needs to solve all by itself?" "Why is it a challenge for a machine to tell the difference between something wet, like a banana peel, and something dry, like paper?"

- Bridge to Design Challenge: Say: "Just like Laila said, we're going to become engineers today! We'll think of ideas, plan, and build the actual mechanism inside our smart bin that uses a sensor and a motor to direct wet and dry waste to the correct compartments. Are you ready to make our bin super smart?"

Possible Strategies for Implementation:

1. Think-Pair-Share: "Sorting Solutions!"

- How to Implement: Following Mrs. Sara's last statement, lead into the next question: "Now that we know the moisture sensor can tell if something is wet or dry, how do you think the bin will move to direct the waste to the correct side? How can the bin have a mechanism that moves to direct the waste?" Have students individually think about this, then pair up and share their initial thoughts on how the sensor and a motor might work together to sort.
- Why it Helps: Encourages individual thought and collaborative brainstorming based on the immediate challenge presented in the dialogue, prompting students to think about the output mechanism after sensor input.

2. Draw the Concept: "Bin's Sorting Secrets!"

- How to Implement: Have students draw a simple diagram of the smart bin. They should try to sketch where the "moisture sensor" (identified in the dialogue) might be placed, where a "motor" would be, and how it might move waste to two different compartments (wet and dry). They can use arrows to show the path of the waste.
- Why it Helps: Visualizes the abstract concepts discussed in the dialogue (sensor input, motor output, sorting paths) and reinforces spatial reasoning for their design.

3. Role-Play Engineering Team: "The Smart Bin Assembly!"

- How to Implement: Create small teams where students take on roles like: "Sensor Expert," "Arduino Coder," "Motor Mechanic," and "Bin Designer." Each group can discuss how they would approach assembling and testing the sorting mechanism that uses a sensor and motor, as discussed in the dialogue.
- Why it Helps: Fosters collaboration, encourages students to think about different aspects of the engineering process, and mimics the collaborative nature of real-world projects, preparing them for the actual build.

4. Predict the Next Step: "What Happens After 'Ready!'?"

- How to Implement: After the discussion and planning, ask students to write or draw what they think the very next steps would be to assemble and test the sorting mechanism in the actual bin, building on the concepts of sensor detection and motor action (e.g., "First, we would connect the moisture sensor to the Arduino...").
- Why it Helps: Promotes forward-thinking, reinforces sequential thinking in engineering, and primes them for the upcoming hands-on build.

Let's Think!



Objective:

Students will develop their understanding of the problems associated with not segregating waste (wet vs. dry). This activity promotes critical thinking about the consequences of improper waste disposal.



List out at least three problems that could happen if food scraps and paper or plastic are all thrown into the same bin.

Handwriting practice lines for listing problems.

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Figure 24: Overflowing Trash Bin.

Possible Answers:

List out at least three problems that could happen if food scraps and paper or plastic are all thrown into the same bin.

- Contamination of Recyclables: Food scraps and wet waste can contaminate paper and plastic, making them unrecyclable and turning valuable resources into landfill waste.
- Unpleasant Odors and Pests: Food waste decomposes, producing foul smells and attracting insects (like flies) and rodents, which creates unsanitary conditions.
- Environmental Harm (Landfills/Pollution): When organic waste (food scraps) goes to landfills, it decomposes without oxygen, producing methane, a potent greenhouse gas that contributes to climate change. Mixing it also makes it harder to compost or incinerate efficiently.

Possible Strategies for implementation:

- Strategy 1: "Problem Brainstorm - The Mixed-Up Bin!"
 - How to Implement: For the task of listing problems, begin with a whole-class brainstorm. Ask: "Imagine a bin where everything just gets tossed in together – banana peels, plastic bottles, paper. What bad things might happen?" Write their ideas on the board. Then, guide them to pick at least three distinct problems to write down.
 - Why it Helps: Activates prior knowledge about waste, encourages critical thinking about consequences, and directly addresses the task of listing problems.
- Strategy 2: "Real-World Impact & Share-Out!"
 - How to Implement: After students have worked on listing problems, facilitate a whole-class discussion. Have a few students share their identified problems and discuss how these issues impact their school, community, or the environment.
 - Why it Helps: Consolidates learning, allows for peer-to-peer explanation, provides immediate feedback, and connects the identified problems to real-world consequences.

Explorer's Toolkit



Teacher Role:

- **Introducer:** Introduce the activity as a hands-on way to "Unlock the secret of wet and dry! See how this circuit uses a moisture sensor to react and move." Explain that students will build a circuit to detect moisture and observe how it controls a servo motor.
- **Facilitator:** Guide students through the steps of assembling the Arduino circuit with the moisture sensor and servo motor, as shown in the provided image. Ensure they understand how to connect each component safely and accurately.
- **Questioner:** Prompt students to think about how the moisture sensor "sees" wet vs. dry, how the Arduino interprets this, and how it then tells the servo motor to move. Ask guiding questions to help them connect their observations to the concepts of sensor input, processing, and motor output.
- **Observer:** Observe students as they conduct the experiment and record their findings, noting their understanding of cause and effect in the electronic circuit.

Expected From Students:

- **Hands-on Construction:** Students will actively participate in building the Arduino circuit, connecting the moisture sensor and servo motor according to the diagram provided.
- **Following Instructions:** Students will follow the step-by-step instructions to assemble and test the circuit.
- **Observation:** Students will carefully watch how the moisture sensor reacts to wet and dry materials and how the servo motor responds to these changes.
- **Developing Understanding of Sensor-Motor Control:** Students will begin to understand how a sensor's input (moisture detection) can be used by a microcontroller (Arduino) to control an output device (servo motor), demonstrating the "reaction and move" principle.

Parts of the books included.

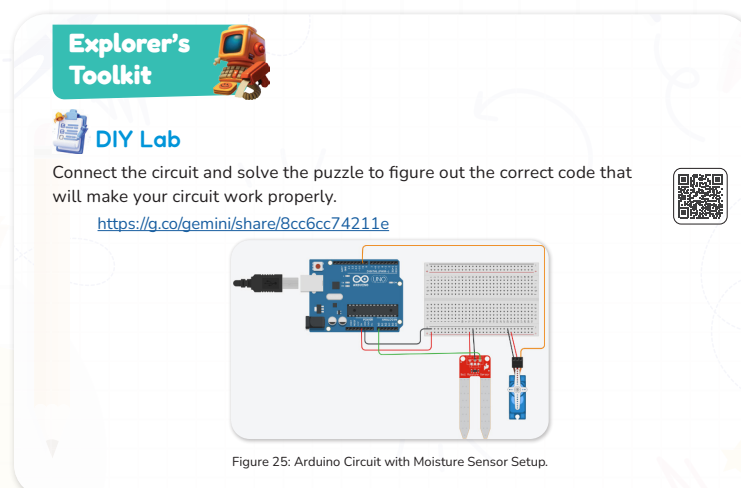


Figure 25: Arduino Circuit with Moisture Sensor Setup.



DIY Lab

Objective:

Students will explore the relationship between moisture levels and a sensor's output, and how this output can be used by an Arduino to control the movement of a servo motor. They will construct a working circuit using a moisture sensor, Arduino, and servo motor, observing and recording how the circuit reacts to wet versus dry conditions. This activity promotes scientific reasoning, observation, and application of engineering principles in a hands-on exploration of smart detection and reaction.

Materials Needed:

- Arduino Uno Board.
- Breadboard
- Soil Moisture Sensor Module.
- Servo Motor
- Jumper Wires (assorted lengths).
- USB Cable (for connecting Arduino to computer).
- Computer with Arduino IDE installed (for code upload).
- Small containers of water.
- Materials to test (a dry piece of paper, a banana peel).

Activity Instructions:

1. Prepare Materials in Advance:

- Have all Arduino components, moisture sensors, servo motors, and wires organized at each group station.
- Ensure computers are ready with Arduino IDE.
- Prepare small amounts of water and dry paper towels/cloths for testing.
- Print and distribute the CER Journal worksheet.

2. Introduce the Activity:

- Gather students and say: "Today, we're going to build a circuit that can actually feel if something is wet or dry, just like we discussed for our smart bin! We'll use a moisture sensor as its 'sense of touch' and make a servo motor react. Your mission is to 'Unlock the secret of wet and dry!' with this circuit."
- Ask: "How do you think our moisture sensor will know if something is wet?" "What will happen to our servo motor when the sensor detects something wet?"
- Demonstrate the Setup (Using the Diagram):
- Use the provided circuit diagram to demonstrate how to connect the Arduino, breadboard, moisture sensor, and servo motor.

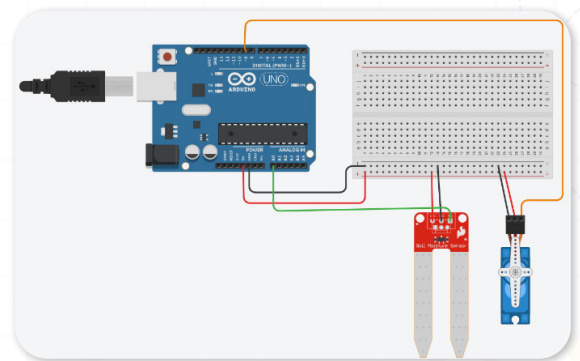


Figure 25: Arduino Circuit with Moisture Sensor Setup.

- Emphasize careful connection of power (VCC/5V), ground (GND), and signal pins for each component.
- Briefly explain that the Arduino acts as the "brain" and wires "send electricity".

Step-by-Step Instructions:

1. Connect the Moisture Sensor:

- VCC → 5V (Arduino).
- GND → GND (Arduino).
- A0 (Analog output) → A0 (Arduino).

2. Connect the Servo Motor:

- VCC → 5V (Arduino).
- GND → GND (Arduino).
- Signal (PWM) → Digital Pin 9 (Arduino).

Upload the Arduino Code: (don't let them write the whole code, prepare another version that includes guide lines or let them play the puzzle to get the code.).

```
#include <Servo.h>
Servo myServo;
int sensorPin = A0;
int servoPin = 9;
int moistureValue = 0;
void setup() {
  myServo.attach(servoPin);
  pinMode(sensorPin, INPUT);
  Serial.begin(9600);
}
void loop() {
  moistureValue = analogRead(sensorPin);
  Serial.println(moistureValue);
  if (moistureValue < 500) {
    myServo.write(90);
  } else {
    myServo.write(0);
  }
  delay(1000);
}
```

3. Test the Circuit:

- Insert the moisture sensor into a dry material (paper, plastic) and observe servo position.
- Insert the sensor into a wet material (wet sponge, banana peel) and observe servo movement.

4. Adjust and Improve:

- Modify the threshold value in the code based on real sensor readings.
- Add a second servo to create separate compartments for wet and dry waste.

Possible Strategies for Implementation:

1. Hands-On Exploration & Testing:

- Students work in pairs or small groups to:
 - Assemble their circuit following the diagram.
 - Upload the code to their Arduino.
 - Test the moisture sensor by touching it to dry paper/cloth, then to wet materials.
 - Observe the servo's movement.
 - Fill out their CER Journal with their findings.

2. Support Students as They Work:

- Move around the room, asking prompts:
 - "What sensor value are you getting when it's dry?"
 - "What happens to the servo when you make the sensor wet?"
 - "Why do you think the servo moves to that specific position when it's wet?"
 - "How is the Arduino making the decision based on the sensor?"

Explain



Teacher Role:

- Concept Introducer: Explain that the video will demonstrate how moisture sensors work, focusing on how they detect wet and dry conditions. Highlight that students will learn how these sensors are used in various applications, including smart systems.
- Observation Guider: Direct students' attention to key aspects in the video, such as how the sensor's probes interact with wet and dry materials, how the sensor's output changes, and any examples of real-world applications.
- Discussion Facilitator: Lead a post-viewing discussion to help students articulate their observations and connect them to the concepts of electrical conductivity, resistance, and sensor applications. Clarify any misconceptions.

- Vocabulary Builder: Introduce and reinforce terms like "moisture sensor," "conductivity," "resistance," "wet," "dry," "analog/digital signal," and potentially any specific terms used in the video ("capacitive sensor," "resistive sensor").

Expected From Students:

- Focused Viewing: Students will watch the video attentively to understand how moisture sensors detect wet and dry conditions.
- Observation Identification: Students will identify and note how the sensor reacts to wet and dry materials, and how this is shown in the video (e.g., changes in a meter reading, a light turning on/off).
- Active Thinking: Students will consider how the ideas in the video relate to the smart bin project and other applications where moisture sensing is important.
- Verbal Sharing: Students will participate in the pre- and post-watching discussions, sharing their initial thoughts and observations from the video.

Parts of the books included.



Screen Lessons

Objectives:

Students will observe how moisture sensors function by watching a video demonstration. This will reinforce their understanding of how sensors detect environmental conditions and provide input for automated systems

Video link: <https://www.youtube.com/watch?v=17PBh9JFKzE>



Possible Strategies for implementation:

1. Pre-Watching: Activate Prior Knowledge & Set a Purpose.

- Strategy: "What Do You Know?" Begin by asking students what they already know about sensors in general. Then, ask if they can think of any devices that might use

moisture sensors (automatic sprinklers, washing machines, soil monitoring systems). This connects the video to their prior knowledge.

2. During Watching: Active Engagement.

- Strategy: "Sensor Tracker." Provide students with a simple chart or ask them to mentally track what happens in the video when the sensor touches a dry material and when it touches a wet material. Encourage them to note any changes they see (a number changing, a light turning on/off).

3. Post-Watching: Discussion and Reflection.

- Strategy: "The Wet & Dry Story." After the video, ask students direct questions about the sensor's behavior: "How did the video show the sensor reacting to something dry?" "How did it show the sensor reacting to something wet?" "What could a computer or Arduino do with that information?" Encourage them to use the terms "moisture sensor," "wet," and "dry" in their answers.



Dig Deeper

Objective:

Students will develop a deeper understanding of analog and digital electronics, recognizing the fundamental differences between these two ways that information is represented and processed in technology. By actively engaging with an article (or resource) like "Analog vs. Digital Electronics: Difference and Comparison," they will practice critical thinking, information gathering, and communication skills as they analyze how devices gather, store, and communicate information, articulate new insights, and see how these concepts are applied in smart systems.

Article link: <https://www.eletimes.com/analog-vs-digital-electronics-difference-and-comparison>



Possible Strategies for implementation:

1. Pre-Reading: Activating Prior Knowledge & Setting Purpose.

Strategy: KWL Chart - "Machine Talk!"

- How to Implement:
 - Introduce the article topic: "Today, we're going to read about the two main ways that machines 'talk' and share information using electricity – called analog and digital! This is super important for understanding how our smart bin and all smart devices work. Before we start, let's think about what we already know and what we want to find out about how machines communicate."
 - Distribute a KWL chart (Know, Want to Know, Learned) or draw one on the board.
 - "K" (Know): Ask students, "What do you already know about how computers or smart devices get information, or how they turn things on/off?" Have them individually fill in the "K" column with their existing knowledge (e.g., "A light switch is either on or off," "My voice is smooth when I talk").
 - "W" (Want to Know): Next, ask, "What do you want to learn or wonder about how machines send signals, or the difference between 'analog' and 'digital'?" Have them

fill in the "W" column.

- Explain that reading the article ("Analog vs. Digital Electronics: Difference and Comparison") will help them fill in the "L" column later.

Please: Print them the KWL Chart.

Chart resources: <https://url-shortener.me/4OWU>



KWL Chart: Your Learning Adventure!

KNOW 

What do you ****already know**** about this topic?

WANT TO KNOW 

What do you ****want to learn**** about this topic?

LEARNED 

What have you ****learned**** after our lesson?

Fill out the "K" and "W" sections before the lesson. Come back to the "L" section after!

1. During Reading: Focused Information Gathering.

Strategy: Guided Reading with Highlighting/Note-Taking - "Analog/Digital Detectives!"

- How to Implement:
 1. Instruct students to read the article ("Analog vs. Digital Electronics: Difference and Comparison") either individually, in pairs, or as a class.
 2. Provide specific prompts to guide their reading. Examples:
 - "As you read, find the main definition of 'analog' and 'digital'."
 - "Look for words that describe analog signals (e.g., smooth, continuous)."
 - "Look for words that describe digital signals (e.g., on/off, steps)."
 - "Find at least two real-world examples of analog and two of digital devices."
 3. Encourage students to highlight or jot down brief notes/keywords related to these prompts directly on a paper or in their digital document.

2. Post-Reading: Consolidating Learning & Reflection.

Strategy: KWL Chart Completion & Discussion - "Electronics Translators Share!"

- How to Implement:
 1. Bring students back together.
 2. "L" (Learned): Ask students to fill in the "L" column of their KWL chart with new information they learned from the article, paying special attention to answering questions from their "W" column.
 3. Share and Discuss: Facilitate a whole-class discussion where students share what they wrote in their "L" column. Ask for examples and have them explain why a device is analog or digital.

4. Reflect: Ask concluding questions: "What was the biggest difference you learned between analog and digital signals?" or "Why is it important for our Arduino (the brain!) to understand both analog and digital signals?"

Elaborate



Teacher Role:

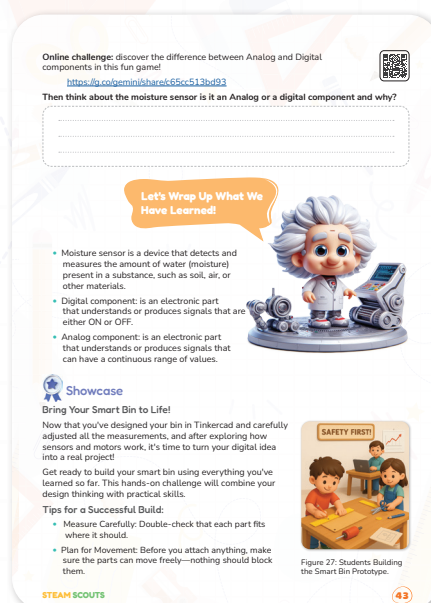
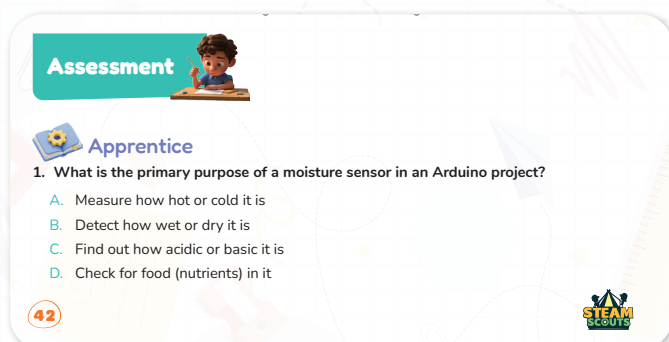
- **Concept Assessor:** Present the multiple-choice question on the primary purpose of a moisture sensor. Guide students in discussing and confirming the correct answer, reinforcing their understanding.
- **Game Facilitator:** Guide students in accessing and playing the "Analog vs. Digital" online challenge game. Encourage them to explore the differences between "smooth talk" Analog and "on/off" Digital components.
- **Discussion Leader:** Lead a focused discussion after the game, prompting students to classify the moisture sensor as either an Analog or a Digital component and explain why, connecting it to the concepts learned in the game.
- **Construction Introducer:** Announce the transition to the "Smart Bin Construction" phase, emphasizing that it's time to "Turn Your Ideas into Reality!"
- **Construction Guide:** Review the "Important Tips for Your Success" for smart bin construction: emphasizing accurate measurement, thinking about movement, safety first, and creativity.
- **Observer & Supporter:** Observe students as they begin the physical construction, offering guidance, ensuring safety, and encouraging their problem-solving approaches.

Expected From Students:

- **Knowledge Application:** Students will answer the multiple-choice question on the moisture sensor's purpose, applying their learned knowledge.
- **Game Engagement:** Students will actively play the "Analog vs. Digital" online challenge game, experimenting to discover the differences between these component types.
- **Critical Classification:** Students will analyze the moisture sensor based on its function and classify it as either an Analog or Digital component, articulating their reasoning.
- **Transition to Hands-on:** Students will actively prepare for the physical construction of their smart bin, understanding the shift from conceptual learning to practical application.
- **Applied Skills:** Students will begin the Smart Bin Construction, focusing on:
 1. Accurate measurement and component fitting.
 2. Considering part movement to avoid obstructions.
 3. Practicing safety with tools and materials.

4. Integrating creative touches into their bin design.

Parts of the books included.



Apprentice

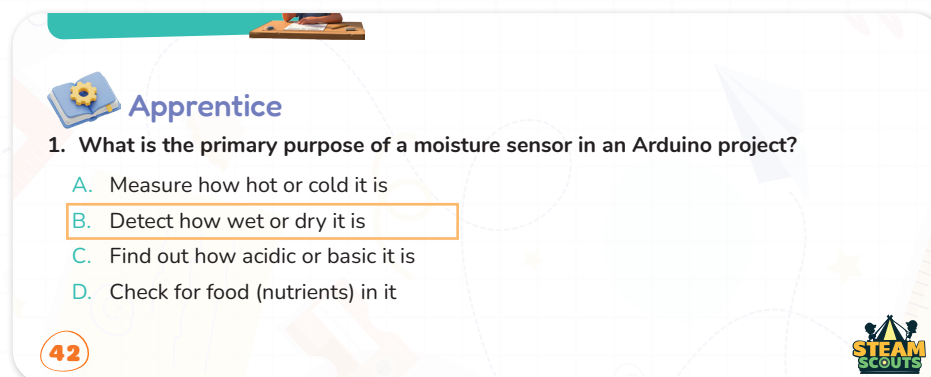
Objective:

Students will develop logical thinking and problem-solving skills by:

- Identifying the primary purpose of a moisture sensor in an Arduino project.
- Experimenting with and understanding the fundamental differences between "smooth talk" Analog components and "on/off" Digital components through an interactive online game.
- Applying this knowledge to classify real-world sensors, such as the moisture sensor, fostering a deeper understanding of how smart systems gather and process information.

Activity link: <https://g.co/gemini/share/c65cc513bd93>

Answers:



Then think about the moisture sensor—is it an Analog or a Digital component, and why?

The moisture sensor is an Analog component because it gives a range of values, not just a yes or no. It measures how much moisture is present in the material and sends a

number (usually between 0 and 1023) to the Arduino. This helps us know how wet or dry something is, instead of just saying "wet" or "dry."

Activity Instructions:

1. **Introduce the Challenge:** Say: "Today, we're going to put our knowledge of sensors and how electronics 'talk' to the test! First, we'll confirm what our moisture sensor is really for, and then we'll play a fun game to truly 'discover the difference between 'smooth talk' Analog and 'on/off' Digital components'. This will help us understand how all our smart bin parts communicate!"
 2. **Answer the Moisture Sensor Question:** Present the multiple-choice question to the class: "What is the primary purpose of a moisture sensor in an Arduino project?"
 3. **Access the Analog/Digital Game:** Direct students to the game link: <https://gemini.google.com/share/c65cc513bd93>
- 
4. **Model First Round (Optional):** Project the game on a screen and play through the first round together. Discuss your thinking aloud (e.g., "This component gives a smooth range of values, so it's probably Analog, not just on/off like Digital.").
 5. **Independent or Partner Play:** Let students play the game alone or in pairs. Encourage verbal reasoning: "Why do you think that's an Analog component?" or "What makes this one Digital?" Allow time for replay and experimentation.
 6. **Reflect as a Group:** Discuss questions like:
 - "Which levels in the Analog/Digital game were tricky? Why?"
 - "What was the biggest difference you noticed between how Analog and Digital components 'talk'?"
 - "Now, let's think about our moisture sensor: Is it an Analog or a Digital component? Why do you think so?" (Guide them to understand it gives a range of values based on moisture, making it Analog).
 - "How does knowing the difference between Analog and Digital help us understand our smart bin better?"

Possible Strategies for Implementation:

- Strategy 1: "Think-Aloud Modeling for Analog/Digital."
 - How to Implement: Teachers demonstrate one or two levels of the Analog/Digital game while talking through their logic. "This component only has two states → on or off → so it must be digital. That's like a light switch!" or "This one change smoothly, like a dimmer switch, so it's analog."
 - Why it Helps: Provides a clear example of the thought process for classifying components and reinforces the core definitions.
- Strategy 2: "Pair and Share: Classifying Our Components."
 - How to Implement: After playing the game, have students work in pairs to discuss

and negotiate their answer to "Is the moisture sensor an Analog or a Digital component and why?" They must agree on a classification and provide a reason together.

- Why it Helps: Encourages collaborative reasoning, deepens understanding through discussion, and prepares them for the whole-class reflection.
- Strategy 3: "Real-World Analog/Digital Hunt."
 - How to Implement: Challenge students to identify one or two other real-world items (besides the smart bin components) that they think use Analog signals and one or two that use Digital signals. (e.g., analog: thermometer, volume knob; digital: light switch, button on a remote).
 - Why it Helps: Connects the abstract concepts to their everyday lives, reinforcing comprehension and application.

Evaluate



Teacher Role:

- Introducer: Introduce and clearly explain the "Bring Your Smart Bin to Life!" activity, marking the transition from theoretical design and component understanding to practical construction.
- Emphasizer: Emphasize the importance of combining digital design thinking with hands-on building skills.
- Tip Reinforcer: Go over and reinforce the "Tips for a Successful Build," ensuring students understand the importance of careful measurement, planning for movement, and safe tool usage.
- Supervisor: Supervise students actively during the construction phase, providing guidance, troubleshooting assistance, and ensuring a safe working environment.
- Motivator: Encourage creativity and personalization in their smart bin projects

Expected From Students:

- Knowledge Applier: Apply their previously acquired knowledge of Tinkercad designs, sensor functions, and motor operations to a physical build.
- Prototyper: Translate their digital bin design into a tangible, working prototype.
- Precise Measurer: Adhere to precise measurements to ensure all parts fit correctly.
- Strategic Planner: Strategically plan the placement of components to allow for free movement and functionality.
- Responsible Tool User: Demonstrate responsible and safe use of all tools and materials.
- Help Seeker: Seek assistance from the teacher when encountering challenges or uncertainties.
- Creative Incorporator: Incorporate creative elements to personalize and enhance their smart bin project

Parts of the books included.

Showcase

Bring Your Smart Bin to Life!

Now that you've designed your bin in Tinkercad and carefully adjusted all the measurements, and after exploring how sensors and motors work, it's time to turn your digital idea into a real project!

Get ready to build your smart bin using everything you've learned so far. This hands-on challenge will combine your design thinking with practical skills.

Tips for a Successful Build:

- **Measure Carefully:** Double-check that each part fits where it should.
- **Plan for Movement:** Before you attach anything, make sure the parts can move freely—nothing should block them.



Figure 27: Students Building the Smart Bin Prototype.

STEAM SCOUTS

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Showcase

Objective:

Students will apply their knowledge of electronics (Arduino as the "brain," IR sensor as "eyes," servo motor to "move parts," and LED as an indicator) to physically construct a functional smart trash bin. Through hands-on building, careful component integration, and creative problem-solving, they will turn their design ideas into a tangible, automated system, demonstrating foundational skills in practical electronics and real-world engineering.

Materials Needed:

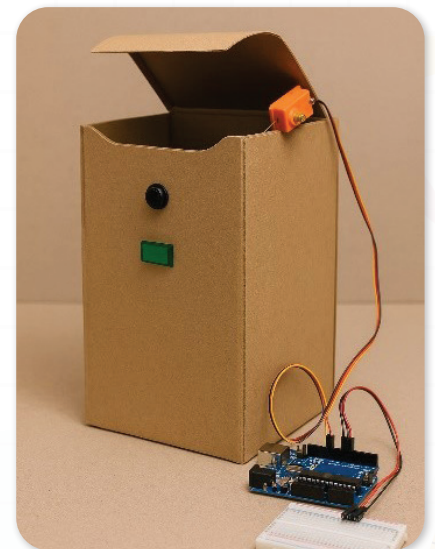
- Cardboard, recycled plastic containers, plastic sheets
- Tools & Mounting Supplies: Scissors, hot glue gun (with adult supervision), strong tape, small zip ties

Activity Instructions:

Safety First! Always handle sharp tools (scissors, craft knives) with extreme caution and tell them to ask for teacher assistance when needed.

Step 1: Plan Your Smart Bin Design & Layout.

- Before you cut or glue anything, decide on the basic shape and size of your smart bin.
- **Think About Movement:** Where will the lid or moving flap be? How will the servo motor attach to make it move smoothly? Sketch your ideas!
- Consider where you will place your components:
 - **IR Sensor:** Where will it best "see" objects (near the opening)?
 - **Servo Motor:** Where can it be hidden but still effectively move the lid/flap?
 - **LED:** Where will it be easily visible as an indicator?
 - **Arduino & Breadboard:** Where can they be safely housed inside the bin, accessible for connections?



- Important Tip: Accurate Measurement! Use rulers to measure and mark your bin materials precisely before cutting. Components need to fit their designated spots!

Step 2: Construct the Bin Body & Lid/Flap.

- Carefully cut and assemble your chosen materials (cardboard, plastic) to form the main body of your trash bin.
- Create a separate lid or a moving flap that your servo motor will eventually control. Make sure it can pivot or slide easily.

Step 3: Integrate the "Eyes" (IR Sensor) & "Indicator Light" (LED).

- Based on your plan, cut a small opening or create a mounting point for your IR sensor on the front or top of your bin, where it can detect objects. Secure it in place (with hot glue or tape).
- Create a small hole or mounting spot for your LED on the bin where it will be easily seen.

Step 4: Mount the "Muscle" (Servo Motor) for Movement.

- Attach the servo motor securely inside your bin. Its position is crucial for controlling the lid or flap. You might need to create a small bracket or use strong tape/glue to fix it.
- Attach the servo's arm to your lid or flap.
 - Important Tip: Think About Movement! Test the physical movement of the lid/flap by hand. Does it open and close freely without obstruction? Does the servo arm have enough range of motion? Adjust its position if needed before final attachment.

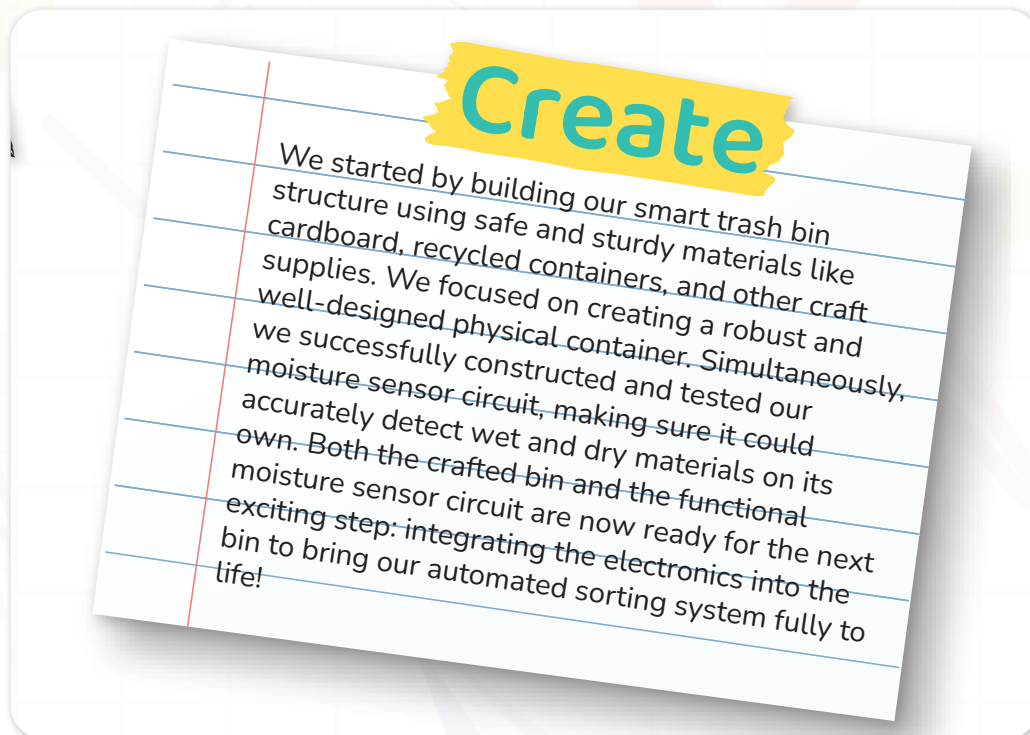
Step 5: Power Up Your Brain (Arduino & Breadboard).

- let them decide the suitable places for each component but don't give it to them

Showcase: Rubric.

Criteria	Excellent (4 Point)	Good (3 Points)	Satisfactory (2 Points)	Needs Improvement (1 Point)
Circuit Assembly	Correct wiring, no errors, all components functional.	Minor wiring issues, mostly functional.	Some errors, needs debugging.	Major wiring issues, not functional.
Code Implementation	Code is correct, optimized, and well-commented.	Code works but needs slight optimization.	Code runs with minor errors.	Code has multiple errors, does not run.
Waste Sorting Logic	Correctly sorts wet and dry waste using sensors.	Works but requires minor adjustments.	Inconsistent sorting.	Sorting does not function correctly.
Construction Quality & Precision.	All physical components are securely and precisely assembled, demonstrating careful measurement and meticulous adherence to the design. The bin is highly sturdy, neat, and well-finished.	Most physical components are securely assembled with good precision; minor imperfections or slight deviations from the design may be present but do not affect overall integrity. The bin is generally sturdy and neat.	Components are partially secured or show noticeable imprecision in assembly, with significant deviations from the design. The bin may be somewhat flimsy or rough in appearance.	Components are poorly or insecurely assembled, leading to a non-functional or unstable bin. Little attention to
Presentation & Explanation	Clearly explains each component and logic.	Explains most parts, but lacks clarity in some areas.	Basic explanation with missing details.	Struggles to explain functionality.
Innovation & Problem Solving	Adds creative features beyond requirements.	Implements basic requirements successfully.	Minimal innovation, meets only basic needs.	Struggles to implement basic logic.

It's time to move to the create part and add this.



Now I Can

Objective:

Students will reflect on their learning by identifying key electronics, programming, and design concepts used while exploring sensors, building circuits, and planning their smart bin. They will demonstrate understanding of how moisture sensors work, how to connect components to an Arduino, the difference between analog and digital signals, and how to apply these concepts in smart device design, preparing them to build functional and intelligent systems.

Activity Instructions

- 1. Review Each (Now I Can) Statement Together:** Read each statement aloud. For each one, ask students to provide examples from their recent activities (like working with the moisture sensor, building circuits, or playing the Analog/Digital game) to reinforce its meaning.
 - For example, for "Describe the main job of a moisture sensor and how it 'tells' us if something is wet or dry," you might ask: "When we tested our moisture sensor, what happened when it touched water? How did it 'tell' us it was wet?"
 - For "Connect the moisture sensor and servo motor to an Arduino board correctly and carefully," you might ask: "What were some important things to remember when we connected the wires to the Arduino and the breadboard?"
 - For "Tell the difference between Analog components... and digital components...", you could ask: "What was a 'smooth talk' component in the game? What was an 'on/off' component? How is our moisture sensor like one of those?"
 - For "Start designing and planning to build my very own smart bin...", you could ask:

"What was one idea you had for where to put a part on your bin? What did you have to think about for it to move correctly?"

- Encourage Self-Assessment: Ask them to thoughtfully check off each skill they feel confident they used and understood during the smart bin project journey.

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **K-2-ETS1-1:** Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool.
 - (Students are addressing the problem of waste management/sorting and designing an improved bin.)
- **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem.
 - (Students are designing and constructing the physical bin and how its components, like the lid and sensor placement, enable its smart functions.)
- **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.
 - (Students test their moisture sensor circuit, and later the integrated bin, to see how well it detects wet/dry or opens/closes, and identify areas for improvement.)

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.4:** Describe people, places, things, and events with relevant details, expressing ideas and feelings clearly.
 - (Students describe their smart bin design, how the sensors work, and how the bin functions.)
- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings.
 - (Students use their bin's physical model, circuit diagrams, and design plans to explain their project.)

3. Common Core State Standards (CCSS) - Mathematics (Measurement & Geometry):

- **1.MD.A.1:** Order three objects by length; compare the lengths of two objects indirectly by using a third object.
 - (Students measure materials for the bin, compare component sizes, and plan placement.)
- **1.G.A.2:** Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) to create a composite shape, and compose new shapes from the composite shapes.
 - (Students use various geometric shapes to construct the bin's body, lid, and internal compartments.)